HEAT REVISION



1. What happens to cause these:

   a. conduction

   b. convection

   c. expansion

   d. contraction



1. Complete the following sentences using the words “convection, conduction, radiation”.

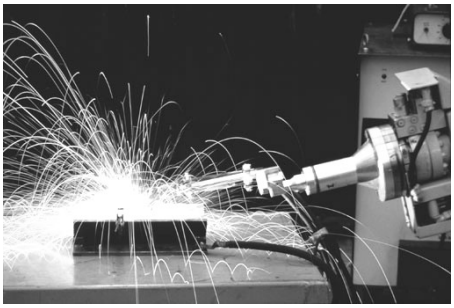
   a. In boiling water the heat is transferred through the water by _____. Heat is transferred through a metal spoon by _____. Heat is transferred to the earth, from the sun, by _____.

2. How do particles move in solids, liquids and gases?

   a. solids:

   b. liquids:

   c. gases:



4. a. Name 3 good conductors:

   b. Name 3 insulators:

   c. Why are insulators often used next to conductors? eg. The plastic handle on a metal saucepan.

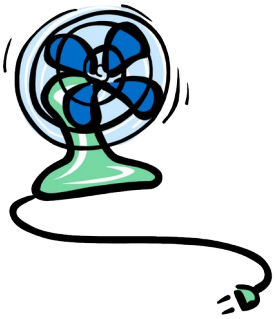
3. Place these colours in order from best absorber to best reflector.

   ‘shiny silver, grey, matt black, gloss (shiny) black, white, orange’

4. How does the design of a thermos flask make it extremely good at slowing down heat loss?

5. What features of an efficient house enable it to save energy?

6. How does a thermometer work?



7. Compare ice, water and steam and explain why these substances have very different properties by referring to the movement of molecules.

8. Explain how a solar hot water system works

9. Give definitions for the following terms:
    a. Freezing
    b. Boiling
    c. Condensing
    d. Subliming
    e. Melting

10. Heat moves around using three methods. What are the names of these three methods and how does each work?

11. The Kimberley has the highest percentage of white cars than anywhere else in Australia. Explain why using your knowledge of heat.

12. Jump on a computer and go to this webpage:

http://www.schoolscience.co.uk/content/3/chemistry/materials/index.html

Read through pages 1 to 9 and then have a go at the quiz questions……..

12. Browse to “Student Resources”, “Science”. “Lower School”, “Year 9”, “Heating & Cooling” and click on the PowerPoint show called “States of Matter”. There will be some questions from this in the test on Friday.